AI-Powered Oral Cancer Detection System

A deep learning approach to early identification and classification of oral lesions, potentially malignant disorders, and oral cavity carcinoma

The Clinical Challenge

Current Limitations

- *Late-stage diagnosis remains common*
- *Visual examination requires extensive expertise*
- *Subtle lesion differences challenge accurate classification*
- *Limited access to specialists in underserved areas*

Our Solution

An intelligent diagnostic system that leverages convolutional neural networks to detect and classify oral lesions across four primary categories with detailed subtype identification.

The model provides risk assessment based on behavioral factors including tobacco use, betel quid consumption, and alcohol intake.

Four-Category Classification System

Oral Cancer (OCA)

Malignant lesions requiring immediate intervention

- *Oral squamous cell carcinoma*
- *Verrucous carcinoma*
- *OCA with leukoplakia*

OPMD (Potentially Malignant)

Pre-cancerous conditions requiring monitoring

- *Leukoplakia (various types)*
- *Erythroplakia*
- *Oral lichen planus (OLP)*
- *Oral submucous fibrosis (OSF)*

Benign Lesions

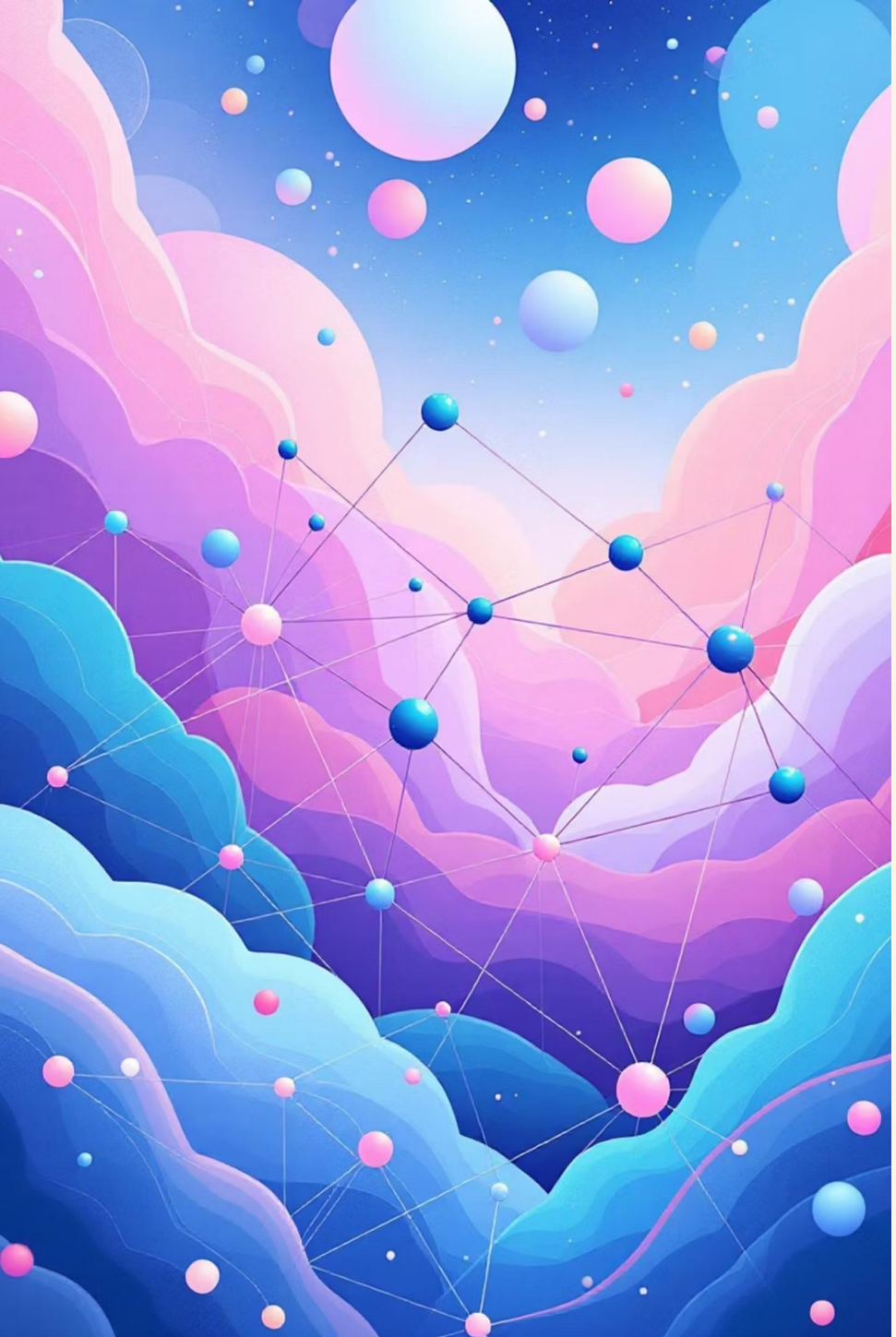
Non-cancerous conditions with favorable prognosis

- *Traumatic ulcers*
- *Candidiasis*
- *Fibrous hyperplasia*
- *Geographic tongue*

Healthy Tissue

Normal oral mucosa with no pathological findings

- *Normal mucosa*
- *Standard tissue variants*



Deep Learning Architecture

01

Data Augmentation

Enhanced training dataset diversity through rotation, flipping, scaling, and color adjustments to improve model robustness

02

CNN Architecture

Convolutional neural networks extract hierarchical features from oral cavity images, identifying patterns invisible to human observers

03

Transfer Learning

Leveraging pre-trained models accelerates training and improves accuracy with limited medical imaging datasets

04

TensorFlow Training

Robust deep learning framework enables efficient model training, validation, and deployment in clinical settings

Data Management & Augmentation

Addressing Data Imbalance

Real-world datasets for oral lesions are often imbalanced, with significantly fewer images available for rare but critical categories like Oral Cancer (129 images) compared to common benign lesions (748 images) or healthy tissue (729 images). This disparity can lead to biased model performance.

Strategic Data Enhancement

To counteract imbalance and improve diagnostic accuracy, we augment minority class images using techniques such as geometric transformations, color variations, and advanced methods like Deep SMOTE. This generates high-quality synthetic data, ensuring comprehensive and unbiased model training.

Dataset

This research utilizes a comprehensive ***dataset of annotated oral cavity images specifically curated for the diagnosis of oral cancer and oral potentially malignant disorders.*** *This dataset is crucial for training and validating our diagnostic models, ensuring they are robust and accurate for real-world applications.*

The dataset was sourced from:

N.S. Piyarathne, S.N. Liyanage, R.M.S.G.K. Rasnayaka, P.V.K.S. Hettiarachchi, G.A.I. Devindi, F.B.A.H. Francis, D.M.D.R. Dissanayake, R.A.N.S. Ranasinghe, M.B.D. Pavithya, I.B. Nawinne, R.G. Ragel, R.D. Jayasinghe. We have obtained proper access and permissions from the dataset owners for its use in this research.

Comprehensive Risk Assessment

Beyond Diagnosis: Behavioral Risk Profiling

Our system doesn't just identify lesions—it evaluates patient-specific risk factors to provide probability assessments for malignancy.

Tobacco & Betel Quid Use

Quantifies exposure risk from chewing tobacco and betel quid consumption patterns

Alcohol Consumption

Integrates drinking history as a multiplicative risk factor

Combined Risk Score

Generates personalized probability estimates for malignant transformation

Research Paper -Reference

Development of an oral cancer detection system through deep learning

Liangbo Li, Cheng Pu, Jingqiao Tao, Liang Zhu, Suixin Hu, Bo Qiao, Lejun Xing, Bo Wei, Chuyan Shi, Peng Chen, and Haizhong Zhang

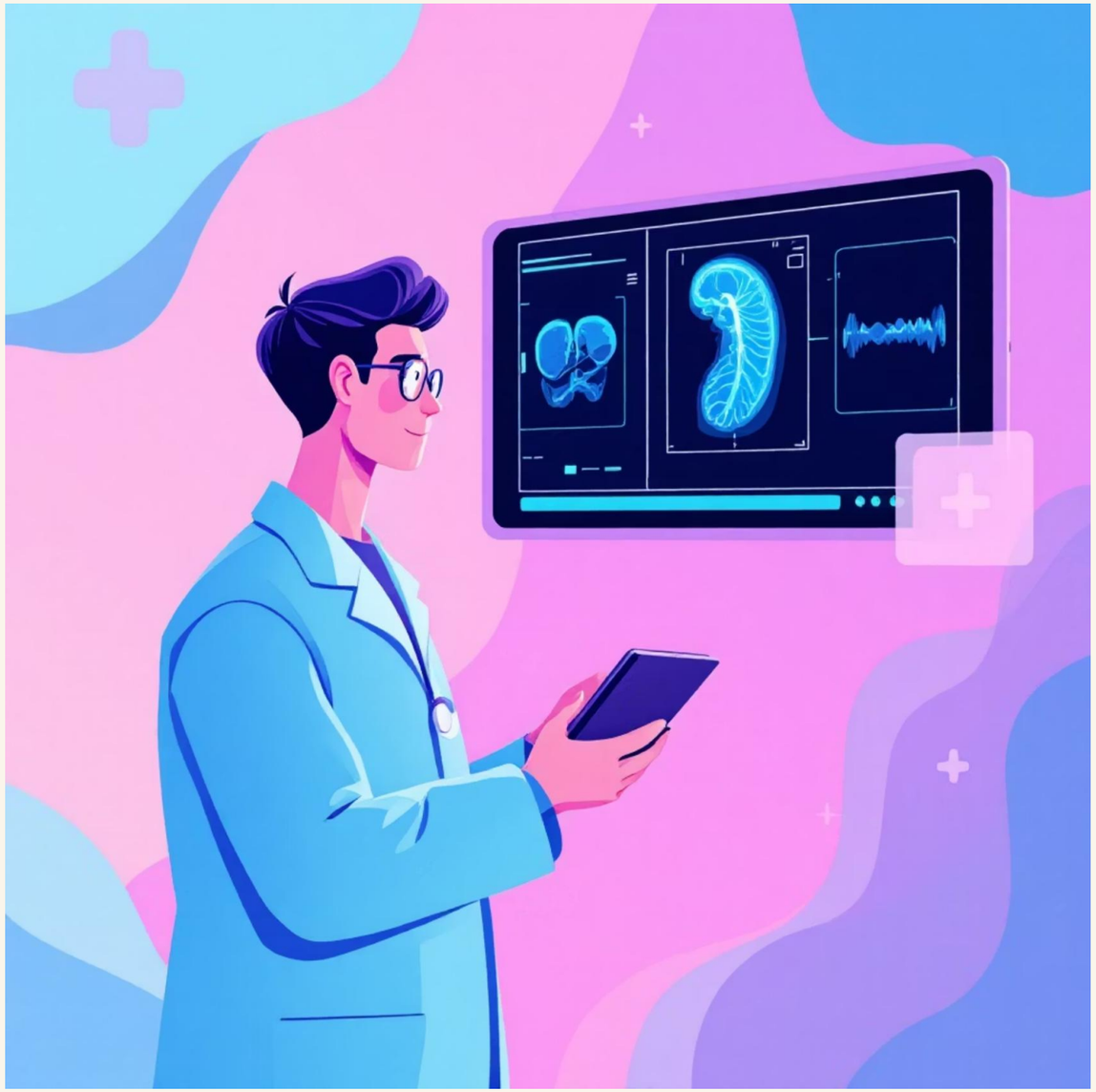
Established foundational methodology for multi-class oral lesion classification using deep learning architectures

Oral Cancer Detection Using Deep Learning Approach"

Ram Kumar Yadav, Priyanka Ujjainkar, Rahul MoriwalDepartment of Computer Science and Engineering & AIG H Raisoni College of Engineering, Nagpur, India

Provided implementation insights for CNN-based diagnostic systems in clinical workflows

Clinical Impact & Future Directions



Transforming Oral Cancer Care

- Early Detection

  Identifying potentially malignant disorders before progression to carcinoma

- Accessible Screening

  Democratizing expert-level diagnosis in resource-limited settings

- Personalized Risk Stratification

  Guiding follow-up intervals and intervention strategies

- Clinical Decision Support

  Augmenting clinician expertise with AI-powered insights

Next Steps: *Prospective clinical validation, integration with electronic health records, and expansion to additional lesion subtypes*